

NICHOLAS TRIGGER

Biomedical Engineer

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SUMMARY

Biomedical Engineering alumnus from Duke University with hands on experience across the medical device design process, from concept and prototyping through verification and validation testing and design history file documentation. Experienced with CAD modeling, test protocol development, and design controls under FDA and ISO 13485 quality systems, with a clinical research background that adds a patient centered perspective to engineering decisions.

EDUCATION

Duke University, Durham, NC

AUG 2022 – MAY 2026

Bachelor of Science in Biomedical Engineering

Relevant Coursework: Biomaterials, Biomechanics, Medical Device Design, Embedded Medical Devices, Imaging Systems

EXPERIENCE

Research and Development Engineer

DEC 2025 – MAY 2026

Resolute Medical (Durham, NC)

- Designed and validated test protocols to support product verification activities and design history file documentation, in alignment with FDA and ISO 13485 requirements.
- Conducted literature reviews and risk-based analysis to inform and refine experimental designs and test protocols.
- Documented and analyzed experimental results, facilitating informed design decisions and technical reporting supporting design history file documentation.

Research and Development Engineer

APR 2025 – MAY 2026

Duke University (Durham, NC)

- Designed, fabricated, and assembled electronic laboratory hardware for deployment across BME teaching labs.
- Developed a stimulus-responsive EKG simulator generating physiologically accurate signals for lab use.
- Validated hardware performance under extended use conditions to ensure durability and safety.

Clinical Research Associate

JUN 2023 – AUG 2024

Christus Spohn Health System (Corpus Christi, TX)

- Supported emergency medicine clinical studies by ensuring compliant data collection and thorough EMR review.
- Collected and documented patient consent in accordance with IRB and FDA guidelines.
- Collaborated effectively with physicians, residents, and research staff to facilitate ongoing clinical investigations.

PROJECTS

Central Line Associated Bloodstream Infection Prevention Device

nicholastrigger.com/projects/clabsi

- Developed handheld UV-C disinfection device for central-line hubs as Duke BME senior capstone, targeting 40,000 annual CLABSIs and eliminating dependency on nurse compliance with Scrub-the-Hub protocol.
- Validated device performance through design verification testing, achieving a 4-log CFU reduction and 2.49 mW optical output targeting the 263-273 nm germicidal window.
- Engineered three custom PCBs, an SLA & FDM 3D printed enclosure, and designed firmware to operate the device safely.

Radial Arterial Line Placement Training Device

nicholastrigger.com/projects/arm/

- Co-developed a durable, anatomically accurate arterial line placement training device to address the shortage of realistic, reusable simulation tools for radial a-line insertion training.
- Iterated through multiple prototype revisions incorporating clinician/user feedback to improve pulse simulation fidelity, tissue realism, and durability under repeated needle insertions.

CERTIFICATIONS AND SKILLS

Certifications: CSSC: Lean Six Sigma White Belt, Nordic Semi: nRF Connect SDK Fundamentals, Matter Fundamentals

Technical Skills: Design Controls, Design History File (DHF), ISO 13485, ISO 14971, Risk Management, FMEA, BioSignal Processing, X-ray, MRI, CT, PET, Medical Device Prototyping, Biomechanics, ISO Standards, FDA Regulations, EMG, Medical Image Processing, Medical Firmware, GD&T, PCB Layout, PCB Design, Python, C, C++, Embedded C, FreeRTOS, Zephyr, JavaScript, TypeScript, HTML/CSS, SQL, SPI, I2C, UART, CAN, BLE, USB, V&V, Bench Testing, System Validation, Bacterial Testing, Phantoms, Data Collection & Analysis, PCB Triage, Prototyping

Tools: SolidWorks, Fusion 360, AutoCAD, OnShape, SolidWorks, KiCAD, EAGLE CAD, Git, GitHub, GitLab, 3D Printing, Laser Cutting, CNC Milling, Manual Machining, Waterjet Cutting, PCB Assembly, PCB Fabrication, Microsoft Office, LaTeX